

Forecasting Intraday Volatility: Evidence from China Gold Futures Market

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Abstract

This study investigates the intraday realized volatility (RV) forecasting of gold futures in China. To predict the RV in the last half an hour before the day trading close (LH), we decompose the whole trading period into several intervals. The empirical results show that RVs in day trading intervals can predict the RV in LH, with RV in ROD interval (from night trading close to the last half an hour before day trading close) having the stronger predictive power. Models which use RVs in separate partitions during ROD deliver better forecasting performance in and out of sample. Meanwhile, RVs in night trading are also informative even after controlling the RVs in day trading. Further, we reconfirm the predictive power of US information from the intraday perspective.

Keywords: China gold futures, Intraday analysis, Night trading, Realized volatility, Volatility forecasting

1. Introduction

As a well-recognized “safe haven”, gold plays a vital role in the global financial system (Baur and Lucey, 2010; Baur and McDermott, 2010; Boubaker et al., 2020). However, in recent years, gold prices have dramatically fluctuated due to turbulence in global economics and politics and the ravages of Covid-19 pandemic (Białkowski et al., 2014; Bams et al., 2017; Guo et al., 2021), which makes forecasts of the gold volatility even more crucial. Meanwhile, gold spots and futures markets in China have experienced rapid growth during the last few years. According to Bloomberg (2021), the Shanghai Gold Exchange (SGE) and Shanghai Futures Exchange (SHFE) rank 4th and 3rd globally in the gold trading volume. Hence, it has great theoretical and practical value to investigate the forecast of volatility in gold related markets in China.

Furthermore, volatility forecasting has always been a key issue in risk management,

asset allocation, derivative pricing and policymaking (Andersen et al., 2005; Khalifa et al., 2011; Wei et al., 2020). Among these, massive studies focus on the forecasting of realized volatility (RV) (Corsi, 2009; Fang et al., 2018; Gong and Lin, 2021; Khalifa et al., 2011; Luo et al., 2016; Lyócsa and Molnár, 2016; Wei et al., 2020). Existing literature highlights the value of the information contained in high-frequency data when predicting daily RV (Gkillas et al., 2020; Jin, 2022; Klein and Todorova, 2021; Luo et al., 2022; McAleer and Medeiros, 2008). But little is known when it refers to intraday forecasting. Current research has confirmed the intraday momentum which results from the hedging demand of the investors (Gao et al., 2018; Baltussen et al., 2021). Given this, we wonder if intraday RV can be predicted.

In this study, we propose the China-based, US-based and China-US-based intraday regression models for the first time to forecast the intraday RV in China gold futures market. Based on the high-frequency gold futures trading data spanning 10 years from SHFE and Commodity Exchange (COMEX), the findings show that (i) intraday RV in the last half an hour before the market close can be significantly predicted by the RVs in other segmented trading time intervals no matter from day or night; (ii) the models using intraday RVs in separate intervals always have better model forecasting performance with positive and higher in- and out-of-sample R^2 ; (iii) adding US gold futures RVs can notably facilitates the performance of RV forecasts in China gold future market.

The contributions of this study are twofold. First, to the best of our knowledge, this paper is the first to conduct the intraday RV forecasting in China gold futures, which fills the research gap in previous literature and provides an important reference for market makers and traders. Second, the research reverifies the predictive power of both the night trading and US information in gold futures market from an intraday perspective rather than interday perspective. Besides, the study of night trading in this paper once again validates the importance of introducing the night trading sessions in China gold futures market.

The remainder of this paper is organized as follows. Section 2 provides methodology. Section 3 describes our data. Section 4 provides empirical results. Finally, Section 5 concludes.

2. Methodology

2.1. Realized variance and intraday trading time intervals

Following Andersen and Bollerslev (1998), we employ the intraday returns to measure the daily realized volatility (RV), which is defined as:

$$RV_t^d = \sqrt{\sum_{i=1}^M r_{t,i}^2} \quad (1)$$

where $r_{t,i}$ refers to the intraday return of i th trade on day t and M is the sampling frequency.

In order to study the predictive power of different intraday time intervals, we divide the entire intraday trading period into day and night and then further segment them into several parts based on the study of Baltussen et al. (2021). As shown in Fig. 1, for day trading, ON denotes overnight; FH denotes the first half an hour after the market open; M denotes middle of the day from the end of FH to 60 minutes before the market close; SLH denotes the second-to-last half an hour before the market close; and LH denotes the last half an hour before the market close. Besides, the combination of the first two parts is marked as ONFH, and the combination of the first four parts is ROD indicating the rest of the day. For night trading, the division of trading time intervals is similar to that of daytime: FHN represents the first half an hour after the market open at night; MN represents middle of the night; and LHN represents the last half an hour before market close at night.

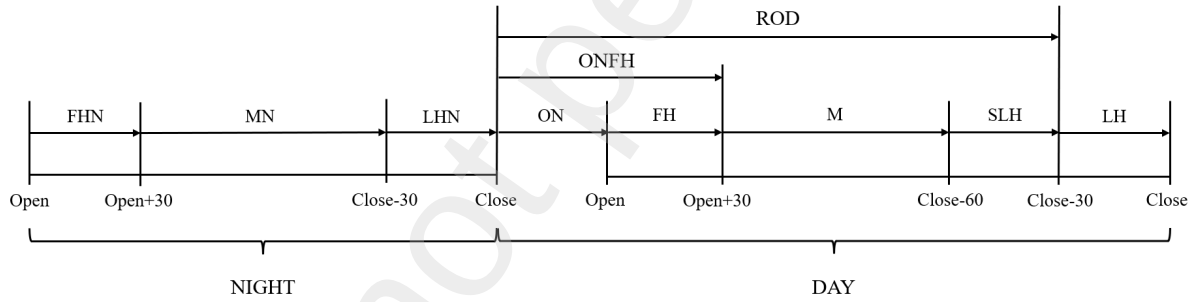


Fig. 1. Intraday trading time intervals.

2.2. China-based intraday models

This paper proposes three new types of intraday regression models regarding the China, US and China-US combined information, respectively. We start with the China-based intraday regression models, which aim to explore the effects of China gold futures information on predicting the China the last half-hour RV.

First, we put forward our basic intraday regression models in terms of day trading (CN-Day models):

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \varepsilon_t \quad (2)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \beta_M^{CN} RV_{M,t}^{CN} + \beta_{SLH}^{CN} RV_{SLH,t}^{CN} + \varepsilon_t \quad (3)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ROD}^{CN} RV_{ROD,t}^{CN} + \varepsilon_t \quad (4)$$

where $RV_{LH,t}^{CN}$, $RV_{ONFH,t}^{CN}$, $RV_{M,t}^{CN}$, $RV_{SLH,t}^{CN}$ and $RV_{ROD,t}^{CN}$ denote China gold futures RVs in LH, ONFH, M, SLH and ROD intervals, accordingly.

Next, the similar night trading models (CN-Night models) are expressed as:

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{FHN}^{CN} RV_{FHN,t}^{CN} + \varepsilon_t \quad (5)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{FHN}^{CN} RV_{FHN,t}^{CN} + \beta_{MN}^{CN} RV_{MN,t}^{CN} + \beta_{LHN}^{CN} RV_{LHN,t}^{CN} + \varepsilon_t \quad (6)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \varepsilon_t \quad (7)$$

where $RV_{FHN,t}^{CN}$, $RV_{MN,t}^{CN}$, $RV_{LHN,t}^{CN}$ and $RV_{NIGHT,t}^{CN}$ represent China gold futures RVs in FHN, MN, LHN and NIGHT intervals, separately.

Further, we include the $RV_{FHN,t}^{CN}$, $RV_{MN,t}^{CN}$, $RV_{LHN,t}^{CN}$ and $RV_{NIGHT,t}^{CN}$ into CN-Day models respectively to examine the predictive power of night information on $RV_{LH,t}^{CN}$ after controlling day information. Thus, we can have CN-Day-FHN models, CN-Day-MN models, CN-Day-LHN models and CN-Day-Night (CN-DN) models with similar equations. Owing to length constraints, this paper takes the CN-DN models as examples:

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \varepsilon_t \quad (8)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \beta_M^{CN} RV_{M,t}^{CN} + \beta_{SLH}^{CN} RV_{SLH,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \varepsilon_t \quad (9)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ROD}^{CN} RV_{ROD,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \varepsilon_t \quad (10)$$

2.3. US-based intraday models

Similarly, we construct the US-based intraday regression models to investigate the predictive power of US gold futures information on China gold futures RV:

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{US} RV_{ONFH,t}^{US} + \varepsilon_t \quad (11)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{US} RV_{ONFH,t}^{US} + \beta_M^{US} RV_{M,t}^{US} + \beta_{SLH}^{US} RV_{SLH,t}^{US} + \varepsilon_t \quad (12)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ROD}^{US} RV_{ROD,t}^{US} + \varepsilon_t \quad (13)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{FHN}^{US} RV_{FHN,t}^{US} + \varepsilon_t \quad (14)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{FHN}^{US} RV_{FHN,t}^{US} + \beta_{MN}^{US} RV_{MN,t}^{US} + \beta_{LHN}^{US} RV_{LHN,t}^{US} + \varepsilon_t \quad (15)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{NIGHT}^{US} RV_{NIGHT,t}^{US} + \varepsilon_t \quad (16)$$

2.4. China-US-based intraday models

Further, this paper focuses on the study of adding US information on the basis of China information. Hence, we introduce the $RV_{ROD,t}^{US}$ and $RV_{NIGHT,t}^{US}$ into our CN-DN

models separately. The China-US-based models considering US day trading information (CN-DN-US-Day models) are:

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \beta_{ROD}^{US} RV_{ROD,t}^{US} + \varepsilon_t \quad (17)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \beta_M^{CN} RV_{M,t}^{CN} + \beta_{SLH}^{CN} RV_{SLH,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \beta_{ROD}^{US} RV_{ROD,t}^{US} + \varepsilon_t \quad (18)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ROD}^{CN} RV_{ROD,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \beta_{ROD}^{US} RV_{ROD,t}^{US} + \varepsilon_t \quad (19)$$

Accordingly, the models considering US night trading information (CN-DN-US-Night models) are:

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \beta_{NIGHT}^{US} RV_{NIGHT,t}^{US} + \varepsilon_t \quad (20)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ONFH}^{CN} RV_{ONFH,t}^{CN} + \beta_M^{CN} RV_{M,t}^{CN} + \beta_{SLH}^{CN} RV_{SLH,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \beta_{NIGHT}^{US} RV_{NIGHT,t}^{US} + \varepsilon_t \quad (21)$$

$$RV_{LH,t}^{CN} = \beta_0 + \beta_{ROD}^{CN} RV_{ROD,t}^{CN} + \beta_{NIGHT}^{CN} RV_{NIGHT,t}^{CN} + \beta_{NIGHT}^{US} RV_{NIGHT,t}^{US} + \varepsilon_t \quad (22)$$

3. Data

In this study, the China and US gold futures price data are obtained from SHFE and COMEX, respectively. We construct 5-min series from the original dataset for our empirical analysis, which can effectively minimize the market microstructure noise (Hansen and Lunde, 2006). The full sample period spans 10 years from July 5, 2013 to January 6, 2022. It should be noted that gold futures in China could only be traded between 9:00 a.m. to 11:30 a.m. and 1:30 p.m. to 3:00 p.m. before July 5, 2013. And since that day, a night trading session was launched, allowing the gold futures in China to be traded during 9:00 p.m. to 2:30 a.m. as well. So, we choose the sample period starting from July 5, 2013. All the aforementioned intraday trading time intervals are based on China Standard Time (CST).

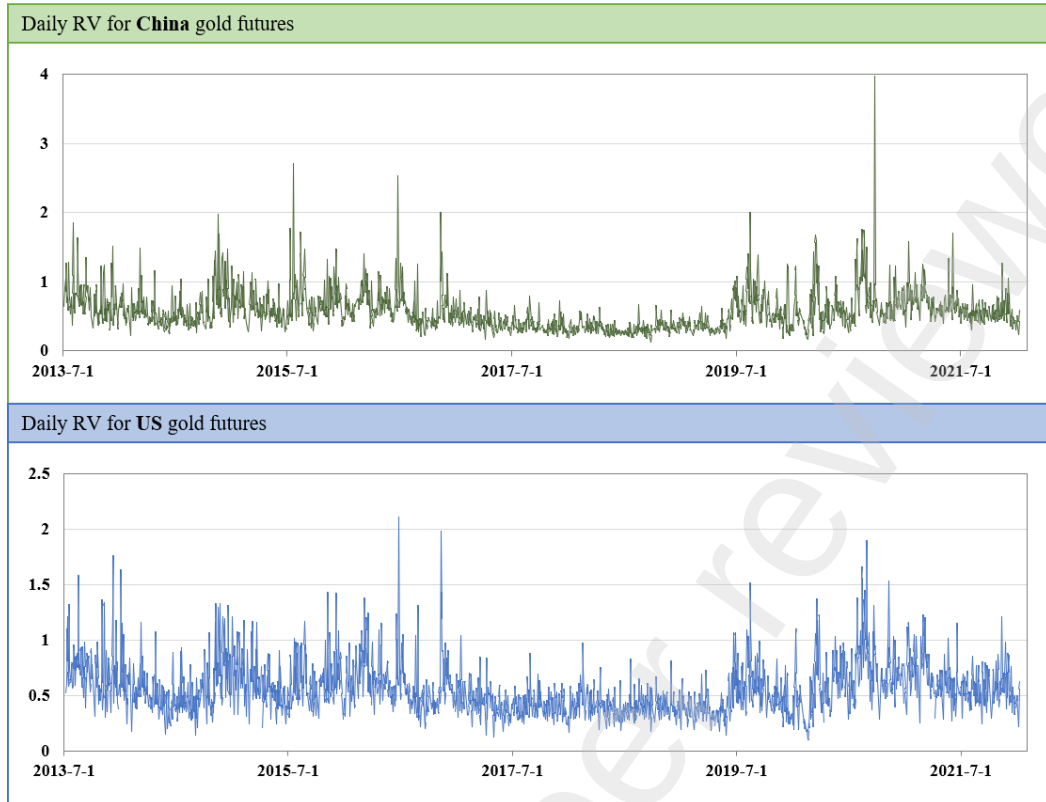


Fig. 2. Time series of daily RV for China and US gold futures.

Fig. 2 plots the time series of daily RV in China and US gold futures markets. It is clearly that the China and US RVs display similar time evolution patterns, which means there are many similarities between these two markets. Meanwhile, the China RVs are generally larger than the US RVs, indicating that China's futures market is more volatile than the US'.

4. Empirical results

In this section, we report the results of the China-based, US-based and China-US-based intraday regression models, respectively. For each model of interest, we also conduct the in-sample and out-of-sample tests. It should be noted that we investigate the out-of-sample performance of models by calculating the out-of-sample R-square (R_{OOS}^2). We set an initial window and update the model parameter when a new observation is available. In addition, we start recursive prediction from July 5, 2018 (5-year samples for the fitting of models) to January 6, 2022 (the last observation). The R_{OOS}^2 is computed as:

$$R_{OOS}^2 = 1 - \frac{\sum_t^T (RV_t - \hat{R}V_t)^2}{\sum_t^T (RV_t - \bar{R}V_t)^2} \quad (23)$$

where RV_t is the true value of RV on day t and $\hat{RV}_t$ is the fitted RV from the predictive models. $\bar{RV}_t$ denotes the historical average RV. A positive R^2_{OOS} indicates that the model makes better forecast.

4.1. Results of China-based intraday models

Here we start with investigating the predictive power of information from China gold futures market. Table 1 shows the sperate results of CN-Day and CN-Night models. It is found that in Panel A, most coefficients are significantly positive with β_{ROD}^{CN} having the largest estimated value (0.404) and t-value (16.90), suggesting that most day trading RVs in China gold futures market have predictive power on RV_{LH}^{CN} and combining all the day trading RVs into the RV_{ROD}^{CN} has the relatively strongest predictive power. As for the forecasting performance of models, the model using RV_{ONFH}^{CN} , RV_M^{CN} and RV_{SLH}^{CN} separately generates positive and the highest in- and out-of-sample R^2 (0.31 and 0.22, respectively), which means that this model has the best forecasting performance.

Table 1

Intraday regression results of the CN-Day and CN-Night models.

Panel A. CN-Day Models				Panel B. CN-Night Models			
β_{ONFH}^{CN}	0.252*** (9.96)	0.034 (1.42)		β_{FHN}^{CN}	0.260*** (10.29)	0.113*** (4.07)	
β_M^{CN}		0.263*** (9.68)		β_{MN}^{CN}		0.273*** (9.57)	
β_{SLH}^{CN}		0.358*** (13.97)		β_{LHN}^{CN}		0.119*** (4.76)	
β_{ROD}^{CN}			0.404*** (16.90)	β_{NIGHT}^{CN}			0.382*** (15.80)
R^2_{adj}	0.06	0.31	0.16	R^2_{adj}	0.07	0.15	0.15
R^2_{OOS}	0.02	0.22	0.13	R^2_{OOS}	0.02	0.22	0.11

Notes: This table provides the in-sample parameter estimation results and out-of-sample R-squares for the China-based intraday regression models over the full sample period from July 5, 2013 to January 6, 2022. The models are based on the information from different intraday trading time intervals in China gold futures market. The day trading time is between 9:00 a.m. to 11:30 a.m. and 1:30 p.m. to 3:00 p.m. (CST). And night trading time is between 9:00 p.m. to 2:30 a.m. All the variables are normalized before regression. ***, ** and * denotes significance at the 1%, 5% and 10% level, respectively.

For the CN-Night models in Panel B, we can gladly find that all the coefficients are significantly positive, which means that all the China night trading RVs predict RV_{LH}^{CN} . This reflects the vital role of night trading information in predicting China RV. While similar to the day trading, RV_{NIGHT}^{CN} has the largest estimated value (0.382) and t-value (15.80) in CN-Night models, revealing that integrating all the night trading RVs into RV_{NIGHT}^{CN} has the relatively strongest predictive power. Besides, the in- and out-of-

sample R^2 s of the model using RV_{FHN}^{CN} , RV_{MN}^{CN} and RV_{LHN}^{CN} separately are the highest as well.

Moreover, we further investigate the combined effects of China day and night trading information on predicting RV_{LH}^{CN} . It can be seen from Panel A in Table 2 that, firstly, the estimates of β_{FHN}^{CN} are always significantly positive, which indicates that the China gold futures RVs in FHN has good predictive power after controlling the day trading information. Secondly, both in- and out-of-sample R^2 s of the models increase after including RV_{FHN}^{CN} . This suggests that compared with CN-Day and CN-Night models which only employ day or night RVs, adding RV_{FHN}^{CN} to our basic CN-Day models facilitates the model forecasting performance. Thirdly, using RV_{ONFH}^{CN} , RV_M^{CN} and RV_{SLH}^{CN} separately also produces the highest in- and out-of-sample R^2 s while RV_{ROD}^{CN} still has the strongest predictive power. The latter finding points out that RV_{ROD}^{CN} contains the relatively most predictive information of all China trading intervals despite day or night.

Table 2

Intraday regression results of the CN-Day-Night models.

Panel A. CN-Day-FHN Models				Panel B. CN-Day-MN Models			
β_{ONFH}^{CN}	0.212*** (8.44)	0.031 (1.28)		β_{ONFH}^{CN}	0.162*** (6.47)	0.017 (0.70)	
β_M^{CN}		0.243*** (8.82)		β_M^{CN}		0.228*** (8.31)	
β_{SLH}^{CN}		0.347*** (13.5)		β_{SLH}^{CN}		0.329*** (12.76)	
β_{ROD}^{CN}			0.361*** (14.81)	β_{ROD}^{CN}			0.314*** (12.54)
β_{FHN}^{CN}	0.221*** (8.83)	0.084*** (3.63)	0.167*** (6.83)	β_{MN}^{CN}	0.313*** (12.46)	0.143*** (5.94)	0.242*** (9.68)
R_{adj}^2	0.11	0.33	0.19	R_{adj}^2	0.15	0.33	0.21
R_{OOS}^2	0.04	0.33	0.20	R_{OOS}^2	0.02	0.34	0.14
Panel C. CN-Day-LHN Models				Panel D. CN-DN Models			
β_{ONFH}^{CN}	0.208*** (7.91)	0.011 (0.46)		β_{ONFH}^{CN}	0.141*** (5.57)	0.008 (0.34)	
β_M^{CN}		0.257*** (9.51)		β_M^{CN}		0.224*** (8.16)	
β_{SLH}^{CN}		0.353*** (13.79)		β_{SLH}^{CN}		0.325*** (12.6)	
β_{ROD}^{CN}			0.377*** (15.00)	β_{ROD}^{CN}			0.297*** (11.65)
β_{LHN}^{CN}	0.141*** (5.36)	0.087*** (3.81)	0.086*** (3.41)	β_{NIGHT}^{CN}	0.335*** (13.23)	0.159*** (6.48)	0.258*** (10.15)
R_{adj}^2	0.11	0.32	0.17	R_{adj}^2	0.16	0.33	0.22
R_{OOS}^2	0.08	0.32	0.19	R_{OOS}^2	0.11	0.35	0.15

Notes: This table provides the in-sample parameter estimation results and out-of-sample R-squares for the China-based intraday regression models over the full sample period from July 5, 2013 to

January 6, 2022. The models are based on the information from different intraday trading time intervals in China gold futures market. The day trading time is between 9:00 a.m. to 11:30 a.m. and 1:30 p.m. to 3:00 p.m. (CST). And night trading time is between 9:00 p.m. to 2:30 a.m. All the variables are normalized before regression. ***, ** and * denotes significance at the 1%, 5% and 10% level, respectively.

The results in Panel B, C and D show similar patterns. It reveals that along with β_{FHN}^{CN} , RV_{MN}^{CN} , RV_{LHN}^{CN} and RV_{NIGHT}^{CN} also have predictive power on RV_{LH}^{CN} and improve the model forecasting performance. Specially, the in- and out-of-sample R^2 s of the model including RV_{NIGHT}^{CN} reach the highest (0.33 and 0.35, respectively), indicating that RV_{NIGHT}^{CN} can best enhance the model forecasting performance. This is why we include RV_{NIGHT}^{CN} as the only China night trading RV in the China-US-based models to conduct the analysis.

4.2. Results of US-based and China-US-based intraday models

In this section, we investigate the predictive power of information from US gold futures market. Table 3 reports sperate results of US-Day and US-Night models. Interestingly, for both day trading and night trading, we can find that all the coefficients are significantly positive. These results tell the fact that US RVs in all trading time intervals have valuable predictive power on the China gold futures RV, which reconfirms the findings of previous literature (Luo et al., 2016; Wei et al., 2020). Moreover, the same as China, the RV_{ROD}^{US} and RV_{NIGHT}^{US} own the relatively strongest predictive power among the models of interest and using segmented RV_{ONFH}^{US} , RV_M^{US} and RV_{SLH}^{US} best improves the model forecasting performance.

Table 3

Intraday regression results of the US-Day and US-Night models.

Panel A. US-Day Models				Panel B. US-Night Models			
β_{ONFH}^{US}	0.222*** (8.72)	0.064** (2.56)		β_{FHN}^{US}	0.208*** (8.15)	0.086*** (3.10)	
β_M^{US}		0.214*** (7.59)		β_{MN}^{US}		0.264*** (9.26)	
β_{SLH}^{US}		0.295*** (10.91)		β_{LHN}^{US}		0.042* (1.65)	
β_{ROD}^{US}			0.336*** (13.62)	β_{NIGHT}^{US}			0.314*** (12.62)
R_{adj}^2	0.05	0.22	0.11	R_{adj}^2	0.04	0.20	0.10
R_{OOS}^2	0.03	0.18	0.04	R_{OOS}^2	0.01	0.18	0.05

Notes: This table provides the in-sample parameter estimation results and out-of-sample R-squares for the US-based intraday regression models over the full sample period from July 5, 2013 to January 6, 2022. The models are based on the information from different intraday trading time intervals in US gold futures market. The trading time is between 9:00 a.m. to 11:30 a.m. and 1:30 p.m. to 3:00 p.m. (CST). All the variables are normalized before regression. ***, ** and * denotes significance at

the 1%, 5% and 10% level, respectively.

So far, we only consider the predictive information from single market. Now we include the US information into our China-based models to see the combined effects on predicting the China gold futures RV. From Panel A of Table 4, we can find that after controlling the information from China market (including both day and night trading), all the estimates of β_{ROD}^{US} are significant and positive, reconfirming the predictive power of US day trading information. Besides, the out-of-sample R^2 of the model involving RV_{ROD}^{US} (0.38) is even higher than that of aforementioned CN-DN models (which have the highest out-of-sample R^2 among all the China-based models). This means that among all the models in this paper, combining information from China and the US especially including RV_{ROD}^{US} into China RV prediction can improve the model forecasting performance to the greatest extent.

Table 4

Intraday regression results of the China-US-based models.

Panel A. CN-DN-US-Day Models				Panel B. CN-DN-US-Night Models			
β_{ONFH}^{CN}	-0.157*** (-3.29)	-0.001 (-0.04)		β_{ONFH}^{CN}	0.117*** (4.62)	0.001 0.04	
β_M^{CN}		0.220*** (7.29)		β_M^{CN}		0.209*** (7.56)	
β_{SLH}^{CN}		0.324*** (12.45)		β_{SLH}^{CN}		0.319*** (12.40)	
β_{ROD}^{CN}			0.416*** (7.53)	β_{ROD}^{CN}			0.270*** (10.48)
β_{NIGHT}^{CN}	0.305*** (12.14)	0.159*** (6.48)	0.255*** (10.04)	β_{NIGHT}^{CN}	0.273*** (10.20)	0.132*** (5.20)	0.214*** (8.05)
β_{ROD}^{US}	0.368*** (7.97)	0.113** (2.26)	0.130** (2.44)	β_{NIGHT}^{US}	0.169*** (6.45)	0.091*** (3.78)	0.134*** (5.19)
R_{adj}^2	0.20	0.30	0.22	R_{adj}^2	0.19	0.34	0.23
R_{OOS}^2	0.15	0.38	0.23	R_{OOS}^2	0.25	0.35	0.26

Notes: This table provides the in-sample parameter estimation results and out-of-sample R-squares for the China-US-based intraday regression models over the full sample period from July 5, 2013 to January 6, 2022. The models are based on the information from different intraday trading time intervals in China and US gold futures market. The trading time is between 9:00 a.m. to 11:30 a.m. and 1:30 p.m. to 3:00 p.m. (CST). All the variables are normalized before regression. ***, ** and * denotes significance at the 1%, 5% and 10% level, respectively.

Panel B of Table 4 shows similar results with all the estimates of β_{NIGHT}^{US} significantly positive and all the R^2 s higher than those of CN-DN models. This provides evidence that the US gold futures RVs in NIGHT also has strong predictive power and can notably improve the model forecasting performance.

5. Conclusion

By considering information from China gold futures market, we forecast gold futures volatility in China particularly based on intraday regressions. It is found that the intraday analysis is equally important. RV in the rest of the day (from previous market close to the last half an hour) can strongly predict the RV in the last half an hour before the market close. And the models which use the information in intraday trading time intervals separately show better model forecasting performance. Meanwhile, we provide evidence that including RVs during night trading time into our basic models based on day information enhance the forecasting performance. Further, we reconfirm the predictive power of US information in gold futures market from the intraday perspective. All the results are robust in the out-of-sample forecasting.

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